

Soft module aspect sizing

Soft module A keeps area six but may reshape between aspect one half and two

Pseudocode

- Soft sizing picks width and height for A under a fixed area and aspect bounds
- Open this module's examples file and find the Pseudocode section
- That written sketch is what you implement on the implement track and what the browser

Algorithm sketch

- Hard golden keeps A at three by two
- Soft teaching pack uses two by three
- Area stays six; deadspace shape changes

Algorithm sketch — try these

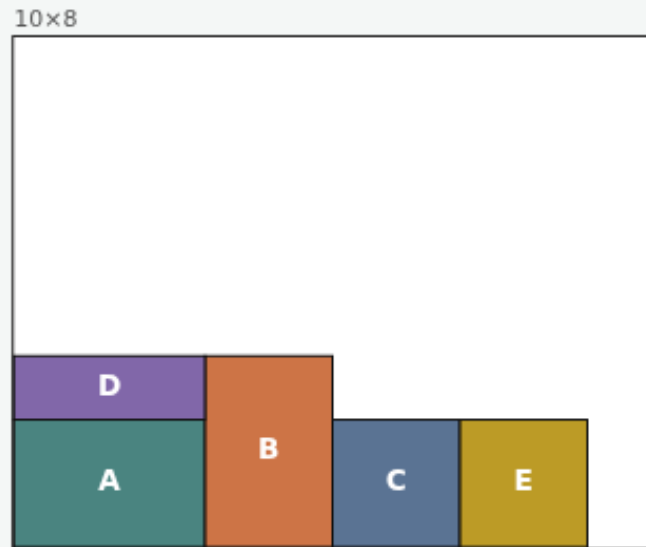
INPUT: soft A area=6, aspect $\in$ [0.5,2]
OUTPUT: (w,h) with $w \cdot h = \text{area}$; pack rest
choose aspect; set w,h; re-pack / legalize
hard modules keep fixed $w \times h$
GOLDEN hard A 3 $\times$ 2 vs soft A 2 $\times$ 3 pack
both area 6; whitespace shape differs

A is soft with area 6

SOFT MODULE SIZING

A is soft with area 6

Step 1 / 5 · soft-a · module03-03-soft-module-sizing



Explanation

Module A is soft: area stays six, but aspect can move between one half and two. Hard modules B through E keep fixed shapes.

- A soft: aspect_min 0.5, max 2
- Area A = $3 \times 2 = 6$
- B-E are hard

A: 3x2 area 6
soft: true

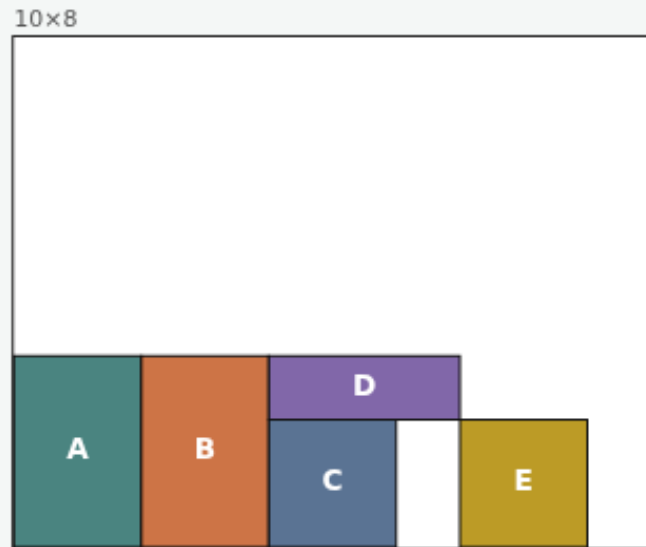
A is soft with area 6

Reshape A to 2×3

SOFT MODULE SIZING

Reshape A to 2×3

Step 2 / 5 · reshape · module03-03-soft-module-sizing



Explanation

Reshape soft A to two by three—still area six. The soft packing moves neighbors so the outline stay legal.

- $2 \times 3 = 6$
- Aspect within $[0.5, 2]$
- Repack around A

A: 2×3
area: 6

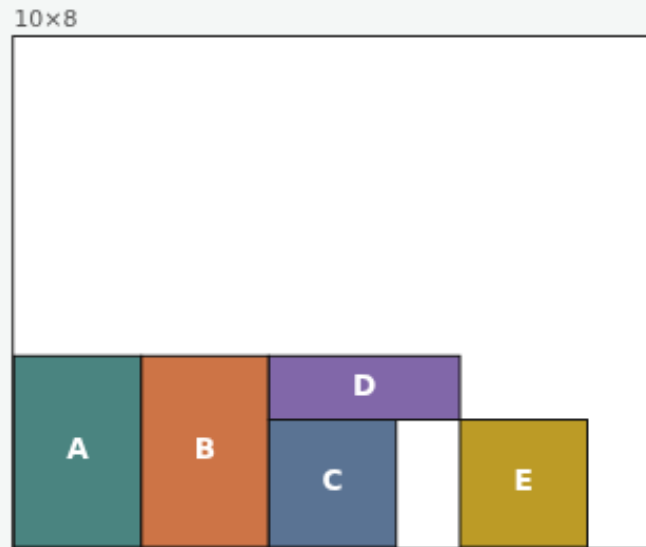
Reshape A to 2×3

Soft packing stays legal

SOFT MODULE SIZING

Soft packing stays legal

Step 3 / 5 · legal · module03-03-soft-module-sizing



Explanation

After reshape, the soft packing remains legal inside ten by eight. Softness is not a license to overflow—legality still gates acceptance.

- legal: true
- Area preserved
- Hard blocks unchanged in size

legal: true

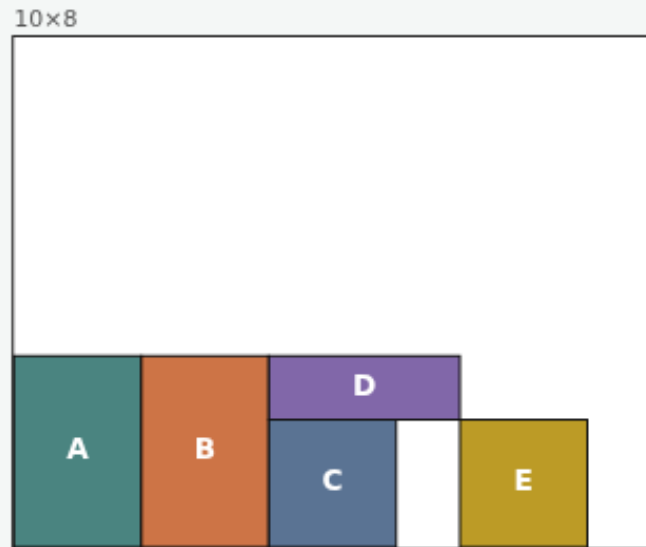
Soft packing stays legal

Hard vs soft views

SOFT MODULE SIZING

Hard vs soft views

Step 4 / 5 · compare · module03-03-soft-module-sizing



Explanation

Compare the hard three-by-two golden with the soft two-by-three packing. Same area budget for A; different whitespace shape.

- Hard A: 3×2
- Soft A: 2×3
- Both legal teaching goldens

hard golden vs soft pack

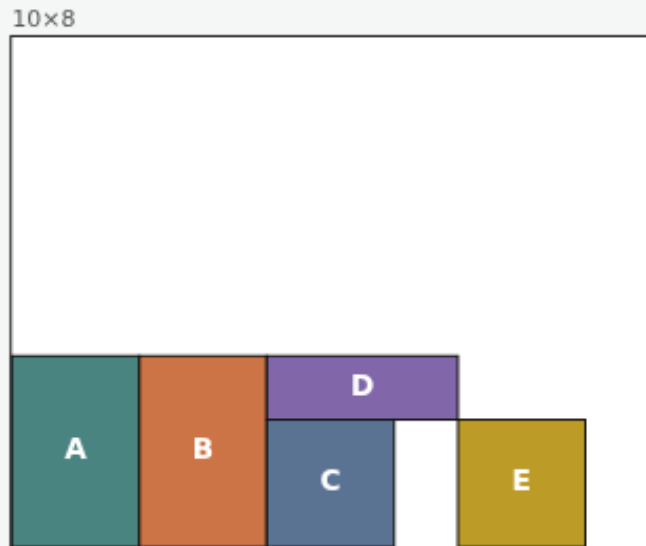
Hard vs soft views

Softness is constrained freedom

SOFT MODULE SIZING

Softness is constrained freedom

Step 5 / 5 · takeaway · module03-03-soft-module-sizing



Explanation

Soft modules trade aspect for packing quality under area and aspect bounds. Always re-check legality after a resize.

- Keep area $\approx$ constant
- Respect aspect_min/max
- Next: fixed macros

starter: A $3 \times 2 \rightarrow 2 \times 3$

Softness is constrained freedom

Browser lab track

- Open soft-module-sizing
- Show hard three-by-two, then Reshape A to two by three
- Confirm area six and legality true on the soft packing

Implement track

- Implement `resize_soft` that preserves area, then pack
- Assert soft A ends at two by three with area six and a legal outline fit

Pitfalls

- Changing area when reshaping; ignoring aspect bounds; leaving hard modules soft by mistake

Your turn

- Accept a legal soft packing with A at two by three
- Next: fix macro D at the origin